

# Statistics Summative Project

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May 2026

## **Abstract**

Placeholder text. This is where the abstract is going to be. I'm going to write it last, as per standard procedure.

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# 1 Hypotheses

We seek to prove that some parts of this data, and some pairs, have association. By performing this test, we attempt to find associations between abstract data values and tests.

$H_0$  = There is no association within the data.

$H_a$  = There is an association, or multiple, within the data.

To reject the null hypothesis, we must prove that the alternative hypothesis is correct. In order to do so, we must use the  $\chi^2$  values and their associated  $p$ -values, as well as graphs and charts depicting these data, to independently analyze them if the  $\chi^2$  tests do not provide enough information to support the alternative hypothesis.

## 1.1 Specific general hypotheses

Some general hypotheses we have about this data are:

- There are associations between philosophical outlook and school.
- There are associations between randomness and age due to maturity.
- MIC will have more STEM-oriented hobbies due to its emphasis on STEM and its attractiveness to STEM enthusiasts.
- SWHS will play more sports on average due to dedicated programs.
- MIC will have a significantly higher proportion of students eating school lunch due to a higher quality of school lunch served.
- There will be a significant difference in weighted GPA between the two schools.
- MIC will have a higher rate of college attendance.
- There will be a significant difference in variability between the two academic environments.

To test these hypotheses, we must perform tests in order to determine their significance, such as two-sample significance tests for means,  $\chi^2$  significance tests, and we will analyze the  $p$ -values at  $\alpha = 0.05$  to determine significance.

## 2 Collection methods

To collect our data, we will use surveys among the student bodies of SWHS and MIC. To achieve this, we will distribute QR codes leading to the survey among the campus; of those, we have around 120.



Figure 1: The QR code used for the survey.

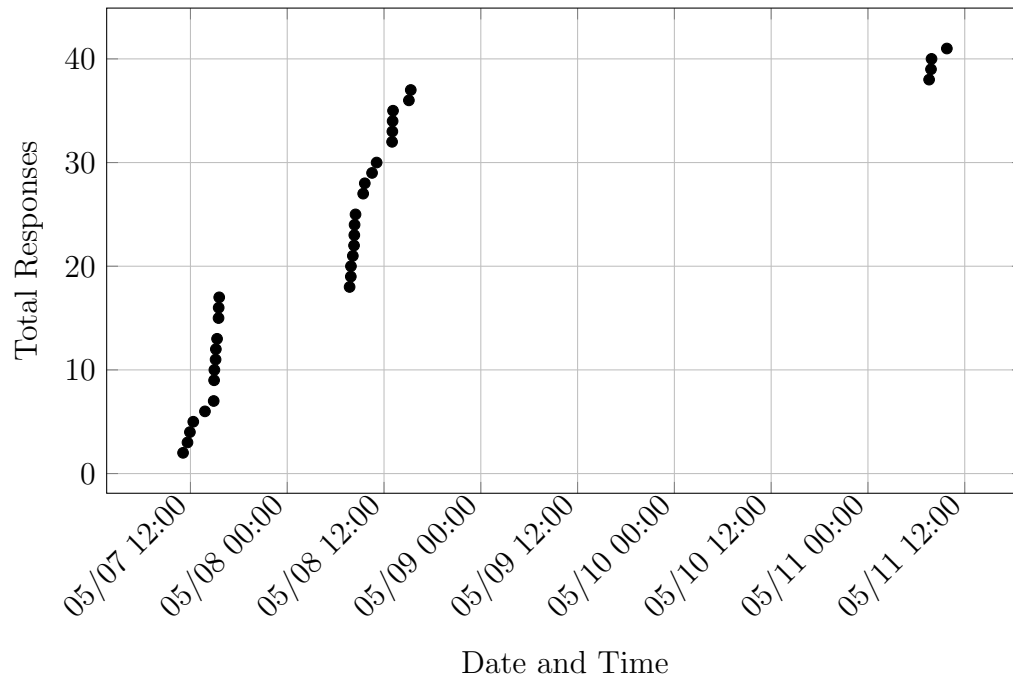
We then contact teachers and administrators in order to further spread our survey among the student population of both SWHS and MIC.

### 2.1 Collection statistics

In total, the survey obtained  $X$  responses out of the possible 1247 students in the population. This is a proportion of  $\hat{p} = 0.X$  for SWHS and  $\hat{p} = 0.X$  for MIC, a — number.

I put people up to the task of handing out QR codes at MIC, and put my full trust in them. The school was in its busy season, so many didn't respond or were too busy to respond.

One small quirk about this collection method is that we get to know when the responses are submitted. This gives us an extra data point as well as enabling us to make graphs such as these:



This graph represents the the total responses in relation to time.

### 3 Appendix: Survey Results

|                 |                 |
|-----------------|-----------------|
| Start time      | 5/7/26 11:49:23 |
| Completion time | 5/7/26 12:21:13 |
| Question I      | G               |
| Question II     | SWHS            |
| Question III    | Fair            |
| Question IV     | 10              |
| Question V      | No;             |
| Question VI     | Fsu             |
| Question VII    | Brown           |
| Question VIII   | Pototoe         |

|                 |                                                               |
|-----------------|---------------------------------------------------------------|
| Question IX     | Superman                                                      |
| Question X      | Idk                                                           |
| Question XI     | Idk                                                           |
| Question XII    | School lunch                                                  |
| Question XIII   | Science;History;Art;Math;English;Music;Foreign<br>Language;   |
| Question XIV    | 3-3.5                                                         |
| Question XV     | Gaming                                                        |
| Question XVI    | 5                                                             |
| Question XVII   | Look at it                                                    |
| Question XVIII  | Lie                                                           |
| Question XIX    | Red                                                           |
| Question XX     | When the height is double the width or more                   |
| Question XXI    | Ask the teacher for an extension on the project               |
| Question XXII   | Rodent                                                        |
| Start time      | 5/7/26 14:46:48                                               |
| Completion time | 5/7/26 14:53:43                                               |
| Question I      | H                                                             |
| Question II     | SWHS                                                          |
| Question III    | Good                                                          |
| Question IV     | 09                                                            |
| Question V      | No;                                                           |
| Question VI     | None                                                          |
| Question VII    | Mr Mitchel                                                    |
| Question VIII   | Pizza                                                         |
| Question IX     | Vanilla                                                       |
| Question X      | GoodKidBand                                                   |
| Question XI     | Gabby                                                         |
| Question XII    | Own lunch                                                     |
| Question XIII   | Music;History;Art;Science;Foreign Lan-<br>guage;English;Math; |
| Question XIV    | 2.5-3                                                         |
| Question XV     | Listening to music                                            |
| Question XVI    | 50                                                            |
| Question XVII   | Look at it                                                    |
| Question XVIII  | Lie                                                           |

|                 |                                                             |
|-----------------|-------------------------------------------------------------|
| Question XIX    | Blue                                                        |
| Question XX     | When the height is the width or more                        |
| Question XXI    | Ask the teacher for an extension on the project             |
| Question XXII   | Goldfish                                                    |
| Start time      | 5/7/26 14:54:38                                             |
| Completion time | 5/7/26 14:58:08                                             |
| Question I      | R                                                           |
| Question II     | SWHS                                                        |
| Question III    | Fair                                                        |
| Question IV     | 09                                                          |
| Question V      | No;                                                         |
| Question VI     | Fsu                                                         |
| Question VII    | Mrs behr                                                    |
| Question VIII   | Raw milk                                                    |
| Question IX     | Chocolate                                                   |
| Question X      | Rebulation                                                  |
| Question XI     | Picasso                                                     |
| Question XII    | School lunch                                                |
| Question XIII   | Science;History;English;Math;Music;Foreign<br>Language;Art; |
| Question XIV    | 3-3.5                                                       |
| Question XV     | Fishing                                                     |
| Question XVI    | 0                                                           |
| Question XVII   | Don't look at it                                            |
| Question XVIII  | Falsely accuse your friend of something worse               |
| Question XIX    | Blue                                                        |
| Question XX     | When the height is double the width or more                 |
| Question XXI    | Ask the teacher for an extension on the project             |
| Question XXII   | Goldfish                                                    |
| Start time      | 5/7/26 14:55:47                                             |
| Completion time | 5/7/26 15:09:43                                             |
| Question I      | F                                                           |
| Question II     | MIC                                                         |
| Question III    | Garbage                                                     |
| Question IV     | 09                                                          |
| Question V      | No;                                                         |



|                 |                                                               |
|-----------------|---------------------------------------------------------------|
| Question VI     | IDK Yet                                                       |
| Question VII    | Ms. Meadows                                                   |
| Question VIII   | Gumbo                                                         |
| Question IX     | Pistachio                                                     |
| Question X      | Built To Spill                                                |
| Question XI     | West Baton Rouge Art Musuem                                   |
| Question XII    | Own lunch                                                     |
| Question XIII   | Science;History;Music;Math;Foreign Lan-<br>guage;English;Art; |
| Question XIV    | 4-4.5                                                         |
| Question XV     | Music                                                         |
| Question XVI    | 53                                                            |
| Question XVII   | Don't look at it                                              |
| Question XVIII  | Lie                                                           |
| Question XIX    | Red                                                           |
| Question XX     | When the height is half the width or more                     |
| Question XXI    | Ask the teacher for an extension on the project               |
| Question XXII   | Rodent                                                        |
| Start time      | 5/7/26 15:13:12                                               |
| Completion time | 5/7/26 15:18:26                                               |
| Question I      | M                                                             |
| Question II     | SWHS                                                          |
| Question III    | Good                                                          |
| Question IV     | 09                                                            |
| Question V      | No;                                                           |
| Question VI     | None                                                          |
| Question VII    | Mrs. Behr                                                     |
| Question VIII   | Burger                                                        |
| Question IX     | Vanilla                                                       |
| Question X      | None                                                          |
| Question XI     | None                                                          |
| Question XII    | School lunch                                                  |
| Question XIII   | Math;Foreign Lan-<br>guage;History;English;Science;Art;Music; |
| Question XIV    | 3-3.5                                                         |
| Question XV     | Games                                                         |

|                 |                                                          |
|-----------------|----------------------------------------------------------|
| Question XVI    | 9                                                        |
| Question XVII   | Warn the teacher about the answer key                    |
| Question XVIII  | Tell the truth                                           |
| Question XIX    | Blue                                                     |
| Question XX     | When the height is half the width or more                |
| Question XXI    | Go to a copy shop                                        |
| Question XXII   | Embrace uncertainty and flee                             |
| Start time      | 5/7/26 15:22:42                                          |
| Completion time | 5/7/26 15:29:17                                          |
| Question I      | K                                                        |
| Question II     | MIC                                                      |
| Question III    | Very good                                                |
| Question IV     | 09                                                       |
| Question V      | No;                                                      |
| Question VI     | None                                                     |
| Question VII    | Ms.Connel                                                |
| Question VIII   | Burger                                                   |
| Question IX     | Vanilla                                                  |
| Question X      | Idk                                                      |
| Question XI     | Idk                                                      |
| Question XII    | School lunch                                             |
| Question XIII   | Science;Music;Foreign<br>guage;History;English;Math;Art; |
| Question XIV    | 3-3.5                                                    |
| Question XV     | Video games                                              |
| Question XVI    | 12                                                       |
| Question XVII   | Look at it                                               |
| Question XVIII  | Lie                                                      |
| Question XIX    | Red                                                      |
| Question XX     | When the height is the width or more                     |
| Question XXI    | Ask the teacher for an extension on the project          |
| Question XXII   | Lizard                                                   |
| Start time      | 5/8/26 7:41:07                                           |
| Completion time | 5/8/26 7:44:23                                           |
| Question I      | M                                                        |
| Question II     | SWHS                                                     |

|                 |                                                             |
|-----------------|-------------------------------------------------------------|
| Question III    | Fair                                                        |
| Question IV     | 11                                                          |
| Question V      | Football;                                                   |
| Question VI     | None                                                        |
| Question VII    | Ms ward                                                     |
| Question VIII   | Alfredo                                                     |
| Question IX     | Cookie dough                                                |
| Question X      | Zach Bryan                                                  |
| Question XI     | Mmg                                                         |
| Question XII    | Own lunch                                                   |
| Question XIII   | History;English;Math;Science;Art;Music;Foreign<br>Language; |
| Question XIV    | 3-3.5                                                       |
| Question XV     | Golf                                                        |
| Question XVI    | 10                                                          |
| Question XVII   | Snatch it                                                   |
| Question XVIII  | Tell the truth                                              |
| Question XIX    | Blue                                                        |
| Question XX     | When the height is double the width or more                 |
| Question XXI    | Ask the teacher for an extension on the project             |
| Question XXII   | Goldfish                                                    |
| Start time      | 5/8/26 7:43:00                                              |
| Completion time | 5/8/26 7:54:26                                              |
| Question I      | H                                                           |
| Question II     | SWHS                                                        |
| Question III    | Good                                                        |
| Question IV     | 11                                                          |
| Question V      | No;                                                         |
| Question VI     | None                                                        |
| Question VII    | Mrs. O'Toole                                                |
| Question VIII   | Pizza                                                       |
| Question IX     | Vanilla                                                     |
| Question X      | Three days grace                                            |
| Question XI     | None                                                        |
| Question XII    | Own lunch                                                   |

|                 |                                                 |      |
|-----------------|-------------------------------------------------|------|
| Question XIII   | History;Foreign                                 | Lan- |
| Question XIV    | guage;Math;Music;Science;English;Art;           |      |
| Question XV     | 2-2.5                                           |      |
| Question XVI    | Playing games                                   |      |
| Question XVII   | 10                                              |      |
| Question XVIII  | Don't look at it                                |      |
| Question XIX    | Tell the truth                                  |      |
| Question XX     | Red                                             |      |
| Question XXI    | When the height is double the width or more     |      |
| Question XXII   | Ask the teacher for an extension on the project |      |
| Start time      | Lizard                                          |      |
| Completion time | 5/8/26 8:05:30                                  |      |
| Question I      | 5/8/26 8:08:06                                  |      |
| Question II     | I                                               |      |
| Question III    | SWHS                                            |      |
| Question IV     | Very good                                       |      |
| Question V      | 10                                              |      |
| Question VI     | Football;                                       |      |
| Question VII    | Idk                                             |      |
| Question VIII   | Coach Rob                                       |      |
| Question IX     | Fries                                           |      |
| Question X      | Chocolate                                       |      |
| Question XI     | Idk                                             |      |
| Question XII    | Idk                                             |      |
| Question XIII   | School lunch                                    |      |
| Question XIV    | Science;Math;History;Foreign                    | Lan- |
| Question XV     | guage;English;Music;Art;                        |      |
| Question XVI    | 4-4.5                                           |      |
| Question XVII   | Gaming                                          |      |
| Question XVIII  | 10                                              |      |
| Question XIX    | Don't look at it                                |      |
| Question XX     | Lie                                             |      |
| Question XXI    | Blue                                            |      |
| Question XXII   | When the height is the width or more            |      |
|                 | Go to a copy shop                               |      |
|                 | Goldfish                                        |      |

|                 |                                                          |
|-----------------|----------------------------------------------------------|
| Start time      | 5/8/26 8:14:47                                           |
| Completion time | 5/8/26 8:17:03                                           |
| Question I      | G                                                        |
| Question II     | SWHS                                                     |
| Question III    | Very good                                                |
| Question IV     | 10                                                       |
| Question V      | No;                                                      |
| Question VI     | I want to be Unemployed                                  |
| Question VII    | MS P                                                     |
| Question VIII   | Steak                                                    |
| Question IX     | Strawberry                                               |
| Question X      | None                                                     |
| Question XI     | None                                                     |
| Question XII    | School lunch                                             |
| Question XIII   | Math;Science;Art;English;Foreign Language;Music;History; |
| Question XIV    | 3.5-4                                                    |
| Question XV     | Work                                                     |
| Question XVI    | 3                                                        |
| Question XVII   | Look at it                                               |
| Question XVIII  | Lie                                                      |
| Question XIX    | Red                                                      |
| Question XX     | When the height is the width or more                     |
| Question XXI    | Ask the teacher for an extension on the project          |
| Question XXII   | Embrace uncertainty and flee                             |
| Start time      | 5/8/26 8:12:04                                           |
| Completion time | 5/8/26 8:19:02                                           |
| Question I      | M                                                        |
| Question II     | SWHS                                                     |
| Question III    | Very good                                                |
| Question IV     | 11                                                       |
| Question V      | No;                                                      |
| Question VI     | Unsure                                                   |
| Question VII    | Mrs. Ward                                                |
| Question VIII   | Crab                                                     |
| Question IX     | Lemon Sorbet                                             |

|                 |                                                             |
|-----------------|-------------------------------------------------------------|
| Question X      | N/A                                                         |
| Question XI     | N/A                                                         |
| Question XII    | Own lunch                                                   |
| Question XIII   | History;English;Science;Art;Music;Math;Foreign<br>Language; |
| Question XIV    | 3-3.5                                                       |
| Question XV     | Video Games                                                 |
| Question XVI    | 10                                                          |
| Question XVII   | Look at it                                                  |
| Question XVIII  | Lie                                                         |
| Question XIX    | Blue                                                        |
| Question XX     | When the height is double the width or more                 |
| Question XXI    | Go to a copy shop                                           |
| Question XXII   | Rodent                                                      |
| Start time      | 5/8/26 9:34:33                                              |
| Completion time | 5/8/26 9:37:53                                              |
| Question I      | S                                                           |
| Question II     | SWHS                                                        |
| Question III    | Very good                                                   |
| Question IV     | 10                                                          |
| Question V      | Soccer;                                                     |
| Question VI     | Cambridge University                                        |
| Question VII    | Ms Kennet                                                   |
| Question VIII   | Pizza                                                       |
| Question IX     | Birthday Cake                                               |
| Question X      | Morgan Wallen                                               |
| Question XI     | Picasso                                                     |
| Question XII    | School lunch                                                |
| Question XIII   | Science;History;Math;Foreign<br>Language;English;Music;Art; |
| Question XIV    | 4-4.5                                                       |
| Question XV     | Sports                                                      |
| Question XVI    | 7                                                           |
| Question XVII   | Don't look at it                                            |
| Question XVIII  | Lie                                                         |
| Question XIX    | Red                                                         |

|                 |                                                               |
|-----------------|---------------------------------------------------------------|
| Question XX     | When the height is double the width or more                   |
| Question XXI    | Ask the teacher for an extension on the project               |
| Question XXII   | Goldfish                                                      |
| Start time      | 5/8/26 7:58:34                                                |
| Completion time | 5/8/26 11:05:49                                               |
| Question I      | I                                                             |
| Question II     | SWHS                                                          |
| Question III    | Garbage                                                       |
| Question IV     | 10                                                            |
| Question V      | Soccer;                                                       |
| Question VI     | Embry Riddle                                                  |
| Question VII    | Teacher Peterson                                              |
| Question VIII   | Sushi                                                         |
| Question IX     | Strawberry                                                    |
| Question X      | The Clash                                                     |
| Question XI     | Piet Mondrian                                                 |
| Question XII    | Own lunch                                                     |
| Question XIII   | Science;Math;History;Art;Music;Foreign Lan-<br>guage;English; |
| Question XIV    | 4.5-5                                                         |
| Question XV     | Cycling                                                       |
| Question XVI    | 15                                                            |
| Question XVII   | Don't look at it                                              |
| Question XVIII  | Lie                                                           |
| Question XIX    | Red                                                           |
| Question XX     | When the height is the width or more                          |
| Question XXI    | Go to a copy shop                                             |
| Question XXII   | Embrace uncertainty and flee                                  |
| Start time      | 5/8/26 12:58:48                                               |
| Completion time | 5/8/26 13:02:52                                               |
| Question I      | J                                                             |
| Question II     | SWHS                                                          |
| Question III    | Fair                                                          |
| Question IV     | 11                                                            |
| Question V      | No;                                                           |
| Question VI     | None                                                          |

|                 |                                                             |      |
|-----------------|-------------------------------------------------------------|------|
| Question VII    | Stevens                                                     |      |
| Question VIII   | Fettuccine Alfredo                                          |      |
| Question IX     | Mint chocolate chip                                         |      |
| Question X      | Huey Lewis                                                  |      |
| Question XI     | Don't know any visual artists                               |      |
| Question XII    | School lunch                                                |      |
| Question XIII   | English;History;Foreign<br>Language;Science;Math;Art;Music; | Lan- |
| Question XIV    | 3.5-4                                                       |      |
| Question XV     | Pass                                                        |      |
| Question XVI    | 13                                                          |      |
| Question XVII   | Look at it                                                  |      |
| Question XVIII  | Lie                                                         |      |
| Question XIX    | Red                                                         |      |
| Question XX     | When the height is double the width or more                 |      |
| Question XXI    | Go to a copy shop                                           |      |
| Question XXII   | Lizard                                                      |      |
| Start time      | 5/8/26 12:57:33                                             |      |
| Completion time | 5/8/26 13:04:26                                             |      |
| Question I      | A                                                           |      |
| Question II     | SWHS                                                        |      |
| Question III    | Good                                                        |      |
| Question IV     | 10                                                          |      |
| Question V      | No;                                                         |      |
| Question VI     | LSU                                                         |      |
| Question VII    | ms bludworth                                                |      |
| Question VIII   | dumplings                                                   |      |
| Question IX     | chocolate                                                   |      |
| Question X      | Lizzy mcalpine                                              |      |
| Question XI     | idk                                                         |      |
| Question XII    | School lunch                                                |      |
| Question XIII   | English;Art;Music;History;Foreign<br>Language;Math;Science; | Lan- |
| Question XIV    | 3-3.5                                                       |      |
| Question XV     | guitar                                                      |      |
| Question XVI    | 4                                                           |      |



|                 |                                                          |
|-----------------|----------------------------------------------------------|
| Question XVII   | Don't look at it                                         |
| Question XVIII  | Lie                                                      |
| Question XIX    | Red                                                      |
| Question XX     | When the height is double the width or more              |
| Question XXI    | Ask the teacher for an extension on the project          |
| Question XXII   | Lizard                                                   |
| Start time      | 5/8/26 14:47:38                                          |
| Completion time | 5/8/26 15:05:41                                          |
| Question I      | R                                                        |
| Question II     | SWHS                                                     |
| Question III    | Good                                                     |
| Question IV     | 11                                                       |
| Question V      | No;                                                      |
| Question VI     | None                                                     |
| Question VII    | Ms. Erwin                                                |
| Question VIII   | Rice with Chicken                                        |
| Question IX     | Mango                                                    |
| Question X      | Ivan Cornejo                                             |
| Question XI     | Frida Kahlo                                              |
| Question XII    | School lunch                                             |
| Question XIII   | Foreign Language;English;Art;History;Math;Science;Music; |
| Question XIV    | 3-3.5                                                    |
| Question XV     | My main hobby is listening to music.                     |
| Question XVI    | 10                                                       |
| Question XVII   | Look at it                                               |
| Question XVIII  | Lie                                                      |
| Question XIX    | Red                                                      |
| Question XX     | When the height is the width or more                     |
| Question XXI    | Go to a copy shop                                        |
| Question XXII   | Goldfish                                                 |
| Start time      | 5/11/26 7:44:07                                          |
| Completion time | 5/11/26 7:49:32                                          |
| Question I      | K                                                        |
| Question II     | SWHS                                                     |
| Question III    | Fair                                                     |
| Question IV     | 11                                                       |

|                 |                                                             |
|-----------------|-------------------------------------------------------------|
| Question V      | No;                                                         |
| Question VI     | UF                                                          |
| Question VII    | Mr. Climer                                                  |
| Question VIII   | Mexican                                                     |
| Question IX     | Lemon                                                       |
| Question X      | Not sure                                                    |
| Question XI     | Not sure                                                    |
| Question XII    | Own lunch                                                   |
| Question XIII   | History;English;Music;Math;Science;Art;Foreign<br>Language; |
| Question XIV    | 3.5-4                                                       |
| Question XV     | Debate                                                      |
| Question XVI    | 140                                                         |
| Question XVII   | Don't look at it                                            |
| Question XVIII  | Lie                                                         |
| Question XIX    | Blue                                                        |
| Question XX     | When the height is the width or more                        |
| Question XXI    | Ask the teacher for an extension on the project             |
| Question XXII   | Lizard                                                      |
| Start time      | 5/11/26 7:46:54                                             |
| Completion time | 5/11/26 7:53:28                                             |
| Question I      | J                                                           |
| Question II     | SWHS                                                        |
| Question III    | Very good                                                   |
| Question IV     | 10                                                          |
| Question V      | Football;                                                   |
| Question VI     | Ucf                                                         |
| Question VII    | Clemens                                                     |
| Question VIII   | Grouper                                                     |
| Question IX     | Vanilla                                                     |
| Question X      | Tyler the creator                                           |
| Question XI     | No clue                                                     |
| Question XII    | School lunch                                                |
| Question XIII   | History;Science;Math;English;Music;Art;Foreign<br>Language; |
| Question XIV    | 3.5-4                                                       |

|                |                                             |
|----------------|---------------------------------------------|
| Question XV    | Photography                                 |
| Question XVI   | 15                                          |
| Question XVII  | Don't look at it                            |
| Question XVIII | Tell the truth                              |
| Question XIX   | Red                                         |
| Question XX    | When the height is double the width or more |
| Question XXI   | Unleash fury on the printer                 |
| Question XXII  | Lizard                                      |